

Practical No. 4

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Aim : Prepare Broad SRS (Software Requirement Specification) for the Project

Smart Education Platform

IEEE 830-based SRS Format

Project Name	Smart Education Platform
Document Type	Software Requirements Specification (SRS)
Standard	IEEE 830-based SRS Format
Technology Used	Python, Django, HTML, CSS, JavaScript, Bootstrap, MySQL
Project Concept	Smart education for efficient, effective, flexible and comfortable digital learning

This SRS describes a proposed educational platform that supports smart education and digital-age learning through online courses, learning resources, assessments, progress tracking, personalized recommendations, and teacher–learner interaction.

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Important terms and sections

1. Introduction

1.1 Purpose

The purpose of the Smart Education Platform is to provide a digital learning environment that enables learners to learn more efficiently, effectively, flexibly and comfortably. The platform supports online learning resources, course management, assessments, progress tracking, personalized learning support and communication between learners and teachers.

- Provide access to digital courses and learning materials.
- Allow learners to study at a convenient time and pace.
- Conduct quizzes, assignments and assessments.
- Track learner progress and academic performance.
- Provide personalized learning recommendations based on learner activity and performance.
- Allow teachers to create and manage courses and educational content.
- Provide administrators with tools to manage users, courses, content and reports.

1.2 Scope

In Scope:

- User registration, login and profile management.
- Course browsing, searching and enrollment.
- Digital learning content such as videos, notes, presentations and links.
- Quiz, assignment and assessment management.
- Automatic evaluation of objective assessments.
- Assignment submission and teacher evaluation.
- Learning progress and performance tracking.
- Personalized learning recommendations.
- Teacher course and content management.
- Administrative management of users, courses and content.
- Learning notifications and announcements.

Out of Scope: The platform does not replace a formal educational institution, teacher, examination authority or professional academic certification process. It provides digital learning support and management.

Benefits: Centralized digital learning, flexible access to educational resources, improved progress monitoring, personalized learning support, convenient assessments and better interaction between learners and teachers.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
DB	Database
API	Application Programming Interface
LMS	Learning Management System
AI	Artificial Intelligence
ML	Machine Learning
Admin	System Administrator
Learner	Student or user who consumes learning content and completes learning activities
Instructor	Teacher or educator who creates and manages learning content

1.4 References

- IEEE 830 — Recommended Practice for Software Requirements Specifications.
- Project concept: Smart Education Platform for digital-age learning.
- Educational platform requirements and proposed application specifications.
- Python, Django, HTML, CSS, JavaScript, Bootstrap and MySQL documentation.

1.5 Overview

This SRS describes the purpose, scope, overall system, users, operating environment, constraints, assumptions, interfaces, functional requirements, performance requirements, database requirements and software quality attributes of the Smart Education Platform.

2. Overall Description

2.1 Product Perspective

The Smart Education Platform consists of a web-based learner and instructor interface connected to a Django/Python backend and a relational database. An administrative interface provides management of users, courses, content, assessments and reports.

High-level flow: Learner / Instructor / Admin → Web Application → Django Backend/API → MySQL Database

2.2 Product Functions

Function	Description
User Management	Registration, login, profile management and role-based access.
Course Management	Create, update, publish and organize courses and topics.
Learning Content	Provide videos, notes, presentations, links and other digital resources.
Course Enrollment	Allow learners to enroll in available courses.
Assessment Management	Create quizzes, assignments and other assessments.
Quiz Evaluation	Automatically evaluate objective questions and display results.
Assignment Management	Allow submission, review, grading and feedback.
Progress Tracking	Record completed lessons, quiz results, assignments and course progress.
Personalized Learning	Recommend suitable resources or topics based on learner activity/performance.
Notifications	Provide course announcements, reminders and important learning updates.
Reports	Provide learner performance and course-level reports to authorized users.

2.3 User Classes and Characteristics

User	Characteristics
Learner / Student	Uses the platform to access courses, study resources, complete assessments and monitor progress.
Teacher / Instructor	Creates courses, manages educational resources, conducts assessments and monitors learners.
Administrator	Manages users, courses, content, permissions, reports and system settings.
Content Manager	Manages or reviews educational resources where this role is enabled.

2.4 Operating Environment

- Client: Modern web browser on desktop, laptop, tablet or smartphone.
- Hardware: Device with display, input capability, storage and internet connectivity.
- Backend: Python and Django running on a web server.
- Database: MySQL relational database.
- Web UI: HTML, CSS, Bootstrap and JavaScript.
- Internet connectivity is required for server-based operations and online learning content.

2.5 Design and Implementation Constraints

- Backend shall be developed using Python and Django.
- MySQL shall be used as the primary relational database.
- The interface shall be responsive for desktop and mobile browsers.
- JavaScript shall be used for required client-side interactions.
- Internet connectivity is required for online platform operations.
- User and academic data shall be handled securely.
- Role-based access shall restrict protected functionality.

2.6 Assumptions and Dependencies

Assumptions: Users have suitable devices and internet access, learners provide valid information, instructors provide accurate educational content, and administrators maintain appropriate user roles and permissions.

Dependencies: Internet connectivity, web server, Django backend, MySQL database, browser compatibility, digital learning content and availability of platform services.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface

- Landing / Home Page
- Registration and Login
- Learner Dashboard
- Instructor Dashboard
- Administrator Dashboard
- User Profile
- Course Listing and Search
- Course Details and Enrollment
- Learning Content Viewer
- Quiz and Assessment Interface
- Assignment Submission
- Progress and Performance Dashboard
- Personalized Recommendations
- Notifications / Announcements
- Course and User Management
- Reports and Analytics

3.1.2 Hardware Interface

The application shall use supported device hardware such as the display, keyboard/touchscreen, storage and network connectivity where required. Optional audio/video hardware may be used when educational content contains multimedia.

3.1.3 Software Interface

The web application shall communicate with the Django backend/API. The backend shall process requests, apply business logic, authenticate users and communicate with the MySQL database.

3.1.4 Communication Interface

The application shall communicate with the server through an internet connection using HTTP/HTTPS-based communication where applicable.

3.2 Functional Requirements

ID	Function	Requirement
FR-01	User Registration	The system shall allow a new learner, instructor or authorized user to create an account, validate required information and store registered user information.
FR-02	User Login	The system shall allow registered users to log in, validate credentials and access features according to their assigned role.
FR-03	Profile Management	The system shall allow users to view and update permitted profile information.
FR-04	Course Browsing	The system shall display available courses with relevant information such as title, description, instructor and learning topics.
FR-05	Course Search	The system shall allow learners to search and filter courses or topics.
FR-06	Course Enrollment	The system shall allow learners to enroll in available courses and add enrolled courses to their learning dashboard.
FR-07	Learning Materials	The system shall allow learners to access available videos, notes, presentations, links and other course resources.
FR-08	Progress Tracking	The system shall record completed lessons, learning activities and course progress for each learner.
FR-09	Quiz Management	The system shall allow instructors to create, modify and publish quizzes.
FR-10	Quiz Attempt	The system shall allow learners to attempt available quizzes and submit their answers.
FR-11	Quiz Evaluation	The system shall automatically evaluate supported objective questions and display results.
FR-12	Assignment Submission	The system shall allow learners to submit assignments before the configured deadline.
FR-13	Assignment Evaluation	The system shall allow instructors to review submissions, provide marks and feedback.
FR-14	Personalized Recommendations	The system shall analyze available learner activity or performance information and provide relevant learning resources or topics.
FR-15	Instructor Management	The system shall allow instructors to create and manage courses, modules and learning content.
FR-16	Learner Performance	The system shall allow instructors to view learner progress, assessment results and performance information.
FR-17	Notifications	The system shall provide authorized course announcements, reminders and learning notifications.
FR-18	Administrator	The administrator shall be able to manage users, roles, courses, content, assessments and system settings.

3.2 Functional Requirements (continued)

3.2.1 Smart / Personalized Learning

- The system shall use available learner activity and performance information to identify relevant learning resources.
- The system shall provide recommendations such as topics, lessons, quizzes or revision resources.
- Recommendations shall be presented as learning support and shall not prevent the learner from accessing other available content.
- The system should allow future integration of machine-learning based recommendation methods.

3.3 Performance Requirements

- Major screens should load within an acceptable time under normal network conditions.
- Database queries should be optimized to avoid unnecessary delays.
- The system should support multiple learners and instructors according to deployed server capacity.
- Course and assessment data should be retrieved efficiently.
- Progress updates should be stored without unnecessary delay.
- The backend should handle simultaneous requests according to available server capacity.
- Search and course filtering should return results efficiently.

3.4 Logical Database Requirements

MySQL shall be used as the primary database.

Table	Purpose
users	Stores account and authentication-related user information
user_profiles	Stores additional profile information
roles	Stores user roles and permissions
courses	Stores course information
course_modules	Stores modules/topics within courses
learning_materials	Stores information about videos, notes, presentations and links
enrollments	Stores learner-course enrollment information
progress	Stores learner lesson and course progress
quizzes	Stores quiz information
questions	Stores quiz questions
quiz_attempts	Stores learner quiz attempts and results
assignments	Stores assignment information
submissions	Stores learner assignment submissions
grades	Stores marks, grades and instructor feedback
recommendations	Stores or records personalized learning recommendations
notifications	Stores announcements and learner notifications
reports	Stores or generates authorized performance/report information

The system shall maintain unique identifiers, relationships between users and learning records, relationships between courses and content, data integrity, valid assessment records and appropriate timestamps.

3.5 Design Constraints

- The application shall follow a modular design.
- The backend should separate application logic from database operations where practical.
- MySQL shall be used for persistent storage.
- Django shall be used for backend web application development.
- Web interfaces shall be responsive.
- Authentication and role-based authorization shall be implemented for protected functionality.
- Sensitive learner and instructor information shall not be unnecessarily exposed.

3.6 Software System Attributes

3.6.1 Security

- Authenticate users before protected information is accessed.
- Restrict administrative and instructor functions to authorized users.
- Protect learner and instructor information from unauthorized access.
- Use secure communication mechanisms where applicable.
- Validate user inputs.
- Apply role-based authorization to protected resources.

3.6.2 Reliability

The system should maintain consistent course, learner and assessment records, minimize data loss and handle invalid inputs and unexpected conditions gracefully.

3.6.3 Availability

The platform should be available whenever the backend, database and required services are operational. Educational resources should remain accessible according to configured course availability.

3.6.4 Maintainability

The system should use modular components so individual features can be modified, tested or extended without significantly affecting unrelated features.

3.6.5 Usability

- Simple navigation.
- Understandable labels and instructions.
- Appropriate feedback after user actions.
- Minimum unnecessary steps.
- Consistent interface.
- Responsive layout for commonly used devices.

3.6.6 Scalability

The system should allow additional courses, learning resources, assessment types, recommendation methods, user roles and educational features to be added in the future.

4. Appendices

Appendix A — Major Modules

- Authentication Module
- User Profile Module
- Course Management Module
- Course Enrollment Module
- Learning Material Module
- Progress Tracking Module
- Quiz Module
- Assignment Module
- Grading and Feedback Module
- Personalized Recommendation Module
- Notification Module
- Reports and Analytics Module
- Administration Module

Appendix B — Technology Stack

Layer	Technology
Web Application / Backend	Python, Django
Database	MySQL
Frontend / Web UI	HTML, CSS, Bootstrap
Client-side Scripting	JavaScript
Smart Recommendation	Python / Machine Learning methods where implemented
Server	Web Server / Application Server
Communication	HTTP/HTTPS/API

Appendix C — High-Level System Architecture

Learner / Instructor / Admin → Web Browser → Django Web Application / API → MySQL Database

The application interface communicates with the Django backend. The backend authenticates users, processes course and learning requests, manages assessments and recommendations, and stores or retrieves information from the MySQL database.

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Term	Section
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Course Enrollment	2.2, 3.2
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